

SOC/L1 - 6.5. NetworkMiner

Task 1 - Room Introduction

NetworkMiner is an open-source Network Forensic Analysis Tool (NFAT). It runs on Windows, Linux, macOS, and FreeBSD, and can be used two ways: as a passive sniffer that fingerprints hosts, sessions, and open ports without sending any traffic, or as a PCAP parser that reassembles files and certificates from captured traffic for offline analysis.

Task 2 - NetworkMiner in Forensics

The ultimate goal of network forensics is to provide sufficient information to detect malicious activities, security breaches, and network anomalies based on the network traffic.

- Context of captured hosts like their IP and MAC, hostnames, or OS information
- List of potential attack indicators or anomalies like traffic spikes or port scans
- Tools or toolkits used to perform the potential attacks like Nmap

Supported Data Types

- Live Traffic
- Traffic Captures
- Log Files

NetworkMiner is capable of processing and handling both packet pictures (PCAP) and live traffic.

Task 3 - What is NetworkMiner?

Capability	Description
Traffic sniffing	It can intercept the traffic, sniff it, and collect and log packets that pass through the network.
Parsing PCAP files	It can parse pcap files and show the content of the packets in detail.
Protocol analysis	It can identify the used protocols from the parsed pcap file.
OS fingerprinting	It can identify the used OS by reading the pcap file. This feature strongly relies on Satori (opens in new tab) and p0f (opens in new tab) .
File Extraction	It can extract images, HTML files and emails from the parsed pcap file.
Credential grabbing	It can extract credentials from the parsed pcap file.
Clear text keyword parsing	It can extract cleartext keywords and strings from the parsed pcap file.

Operating Modes: There are two main operating modes:

- **Sniffer Mode:** NetworkMiner includes a basic Windows-only sniffing feature, but it is primarily a network forensics tool and is less reliable than dedicated sniffers like Wireshark or tcpdump.
- **Packet Parsing/Processing:** Quickly analyzes packet captures to extract key information and identify initial findings before deeper investigation.

Pros

- OS fingerprinting
- Easy file extraction
- Credential grabbing
- Clear text keyword parsing

- Overall overview

Cons

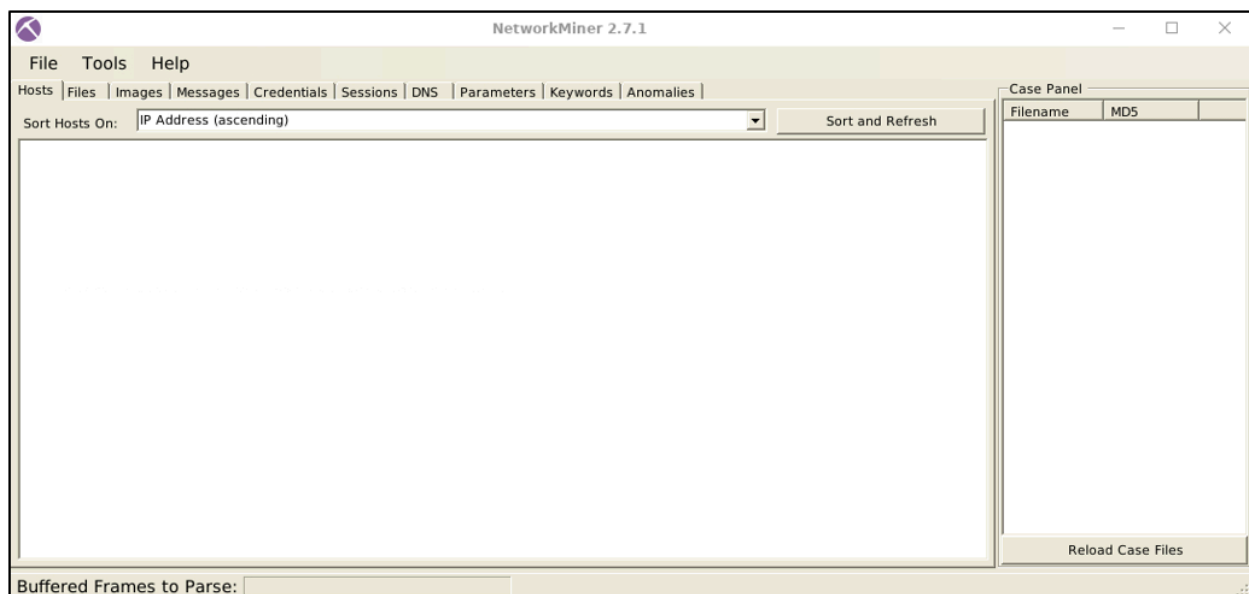
- Not useful in active sniffing
- Not useful for large pcap investigation
- Limited filtering
- Not built for manual traffic investigation

Differences Between Wireshark and NetworkMiner

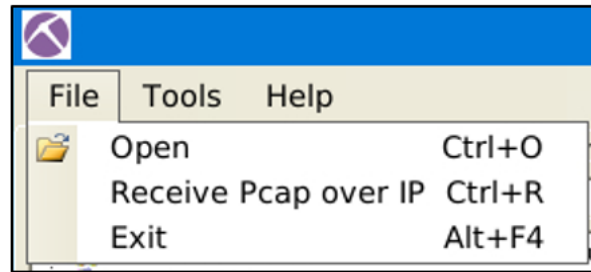
Feature	NetworkMiner	Wireshark
Purpose	Quick overview, traffic mapping, and data extraction	In-Depth analysis
GUI	✓	✓
Sniffing	✓	✓
Handling PCAPS	✓	✓
OS Fingerprinting	✓	✗
Parameter/Keyword Discovery	✓	Manual
Credential Discovery	✓	✓
File Extraction	✓	✓
Filtering Options	Limited	✓
Packet Decoding	Limited	✓
Protocol Analysis	✗	✓
Payload Analysis	✗	✓
Statistical Analysis	✗	✓
Cross-Platform Support	✓	✓
Host Categorisation	✓	✗
Ease of Management	✓	✓

Task 4 - Tool Overview 1

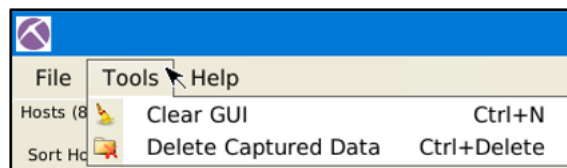
Landing Page: This is the landing page of the NetworkMiner.



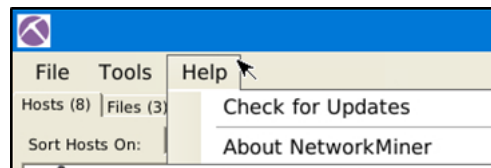
File Menu: Load or drag-and-drop PCAP files for quick traffic analysis; receiving PCAPs over IP is supported but not covered here.



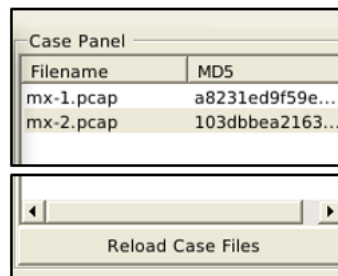
Tools Menu: The tools menu helps you clear the dashboard and remove the captured data.



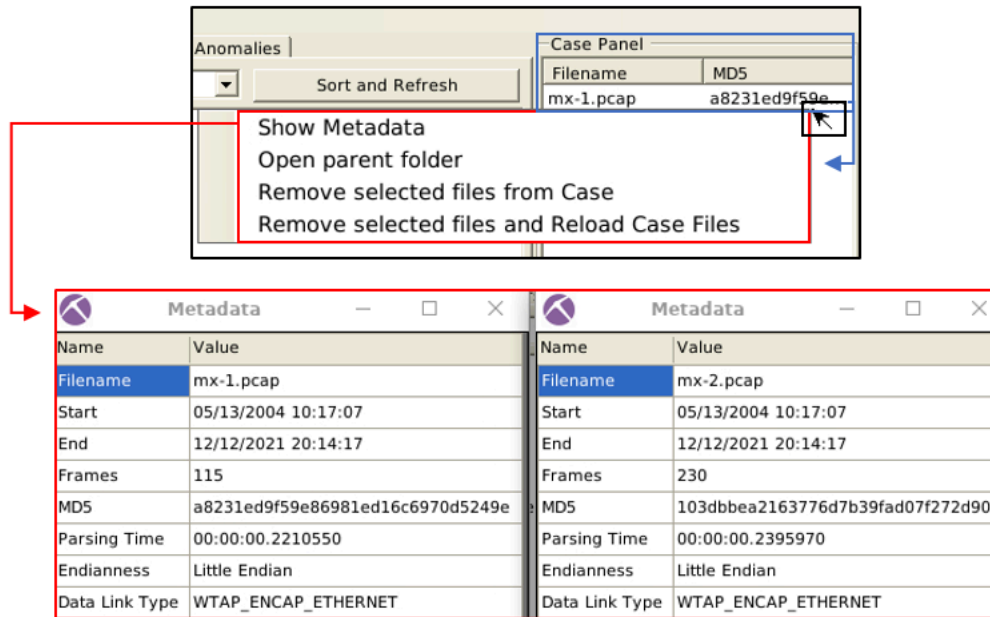
Help Menu: The help menu provides information on updates and the current version.



Case Panel: The case panel shows the list of the investigated pcap files.

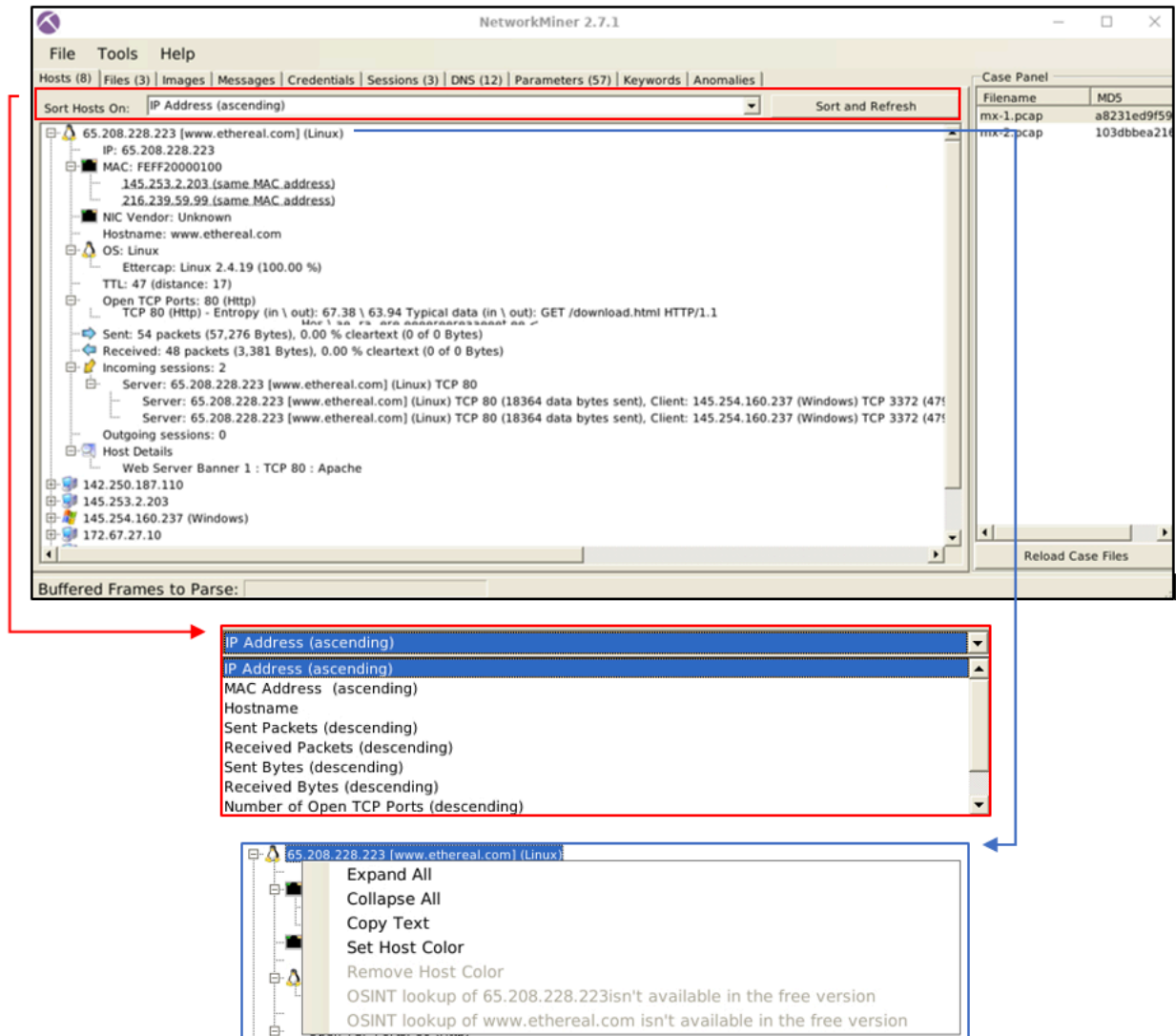


Viewing metadata of loaded files: Open the PCAP, right-click the filename in the Case Panel, and select "**Show Metadata**" to view file details.



Hosts: The "hosts" menu shows the identified hosts in the pcap file. This section provides information on:

- IP address
- MAC address
- OS type
- Open ports
- Sent/Received packets
- Incoming/Outgoing sessions
- Host details



Sessions: The session menu shows detected sessions in the pcap file. This section provides information on:

- Frame number
- Client and server address
- Source and destination port
- Protocol
- Start time

The screenshot shows the 'Sessions' menu in NetworkMiner 2.7.1. The menu displays a table of detected sessions. The table has columns for Frame nr., Client host, C. port, Server host, S. port, Protocol (application layer), and Start time. The sessions are listed as follows:

Frame nr.	Client host	C. port	Server host	S. port	Protocol (application layer)	Start time
1	145.254.160.237 (Windows)	3372	65.208.228.223 [www.ethereal.com] (Linux)	80	Http	2004-05-13
18	145.254.160.237 (Windows)	3371	216.239.59.99 [pagead.google.akadns.net] (pag...	80	Http	2004-05-13
117	145.254.160.237 (Windows)	3372	65.208.228.223 [www.ethereal.com] (Linux)	80	Http	2004-05-13

It is possible to filter specific columns of the session menu as well. This menu accepts four types of inputs;

- "ExactPhrase"
- "AllWords"
- "AnyWord"
- "RegExe"

DNS: This section provides information on:

- Frame number
- Timestamp
- Client and server
- Source and destination port
- IP TTL
- DNS time
- Transaction ID and type
- DNS query and answer
- Alexa Top 1M

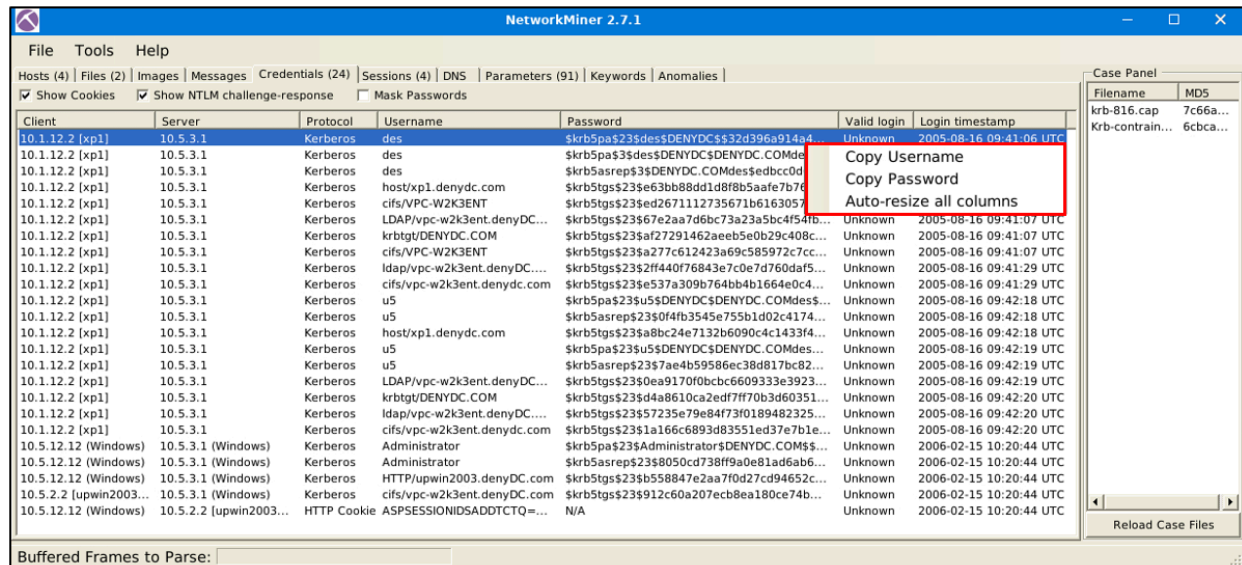
Fr...	Timestamp	Client	Client Port	Server	Server ...	IP TTL	DNS TTL (time)	Transactio...	Type	DNS Query	DNS Answer	Alexa Top 1M
17	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	13:25:21	0x0023	0x0005 (CNAME)	pagead2.googleadsyndication.com	pagead2.google.com	N/A (Pro version only)
17	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:02	0x0023	0x0005 (CNAME)	pagead2.google.com	pagead.google.akadns.net	N/A (Pro version only)
17	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:03	0x0023	0x0001 (HostAd...	pagead.google.akadns.net	216.239.59.104	N/A (Pro version only)
148	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	13:25:21	0x0023	0x0005 (CNAME)	pagead2.googleadsyndication.com	pagead2.google.com	N/A (Pro version only)
148	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:02	0x0023	0x0005 (CNAME)	pagead2.google.com	pagead.google.akadns.net	N/A (Pro version only)
148	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:03	0x0023	0x0001 (HostAd...	pagead.google.akadns.net	216.239.59.104	N/A (Pro version only)
148	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:03	0x0023	0x0001 (HostAd...	pagead.google.akadns.net	216.239.59.99	N/A (Pro version only)
149	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	13:25:21	0x0023	0x0005 (CNAME)	pagead2.googleadsyndication.com	pagead2.google.com	N/A (Pro version only)
149	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:02	0x0023	0x0005 (CNAME)	pagead2.google.com	pagead.google.akadns.net	N/A (Pro version only)
149	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:03	0x0023	0x0001 (HostAd...	pagead.google.akadns.net	216.239.59.104	N/A (Pro version only)
149	2004-05-13 10:1...	145.254.160.237 ...	3009	145.253...	53	249	00:02:03	0x0023	0x0001 (HostAd...	pagead.google.akadns.net	216.239.59.99	N/A (Pro version only)

Filename	MD5
mx-1.pcap	a8231ed9f59e...
mx-2.pcap	103dbbea2163...

Credential: The credentials menu shows extracted credentials and password hashes from investigated pcaps. NetworkMiner can extract credentials including:

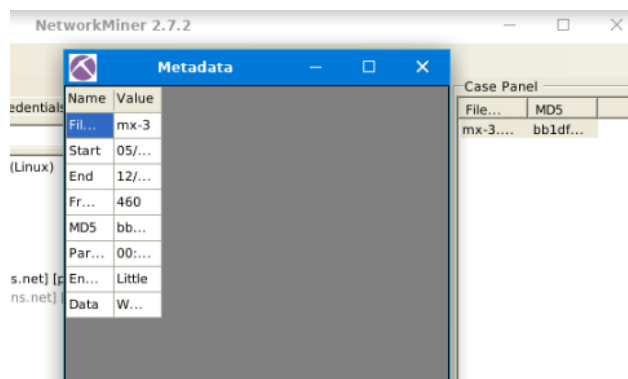
- Kerberos hashes
- NTLM hashes
- RDP cookies
- HTTP cookies
- HTTP requests
- IMAP
- FTP
- SMTP
- MS SQL

The right-click menu is helpful in this part as well. You can easily copy the username and password values.



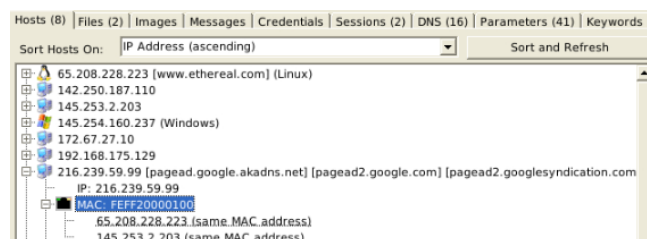
Answer the questions below

Use the "~/Desktop/Exercise Files/mx-3.pcap" file to answer the questions below. What is the total number of frames? (Right-click the case name in the Case panel, then select **Show Metadata**.) 460



How many IP addresses use the same MAC address with host 145.253.2.203?

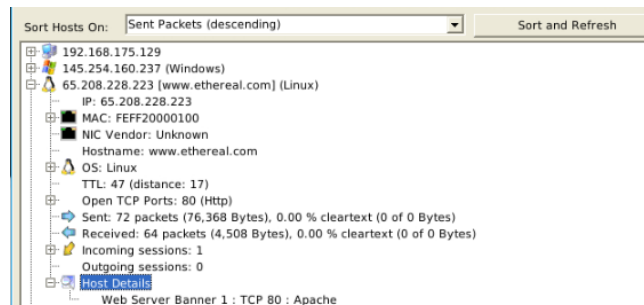
2 (Go to the last IP address, expand it, and select **MAC Address**. Two MAC addresses will be displayed.)



How many packets were sent from host 65.208.228.223? 72 (choose ip)



What is the name of the webserver banner under host 5.208.228.223? Apache



Use the "~/Desktop/Exercise Files/mx-4.pcap" file to answer the questions below. What is the extracted username for the 02694W-WIN10 host?

#B\Administrator (click credentials)

Anomalies				
Hosts (10)	Files (4)	Images	Messages	Credentials (2)
☒ Show Cookies ☒ Show NTLM challenge-response ☐ Mask Passwords				
Client	Server	Protocol	Username	Password
172.16.66.37 [02694W-WIN10]	172.16.66.36	NTLMSSP	#B\Administrator	NTLM Challenge: 136B077D942D9...
172.16.66.37 [02694W-WIN10]	172.16.66.36	NTLMSSP	#B\Administrator	\$NETNTLMv2\$#B\$136B077D942D9...

What is the extracted password for the user logged into the 02694W-WIN10 host? Enter the full NTLM hash.

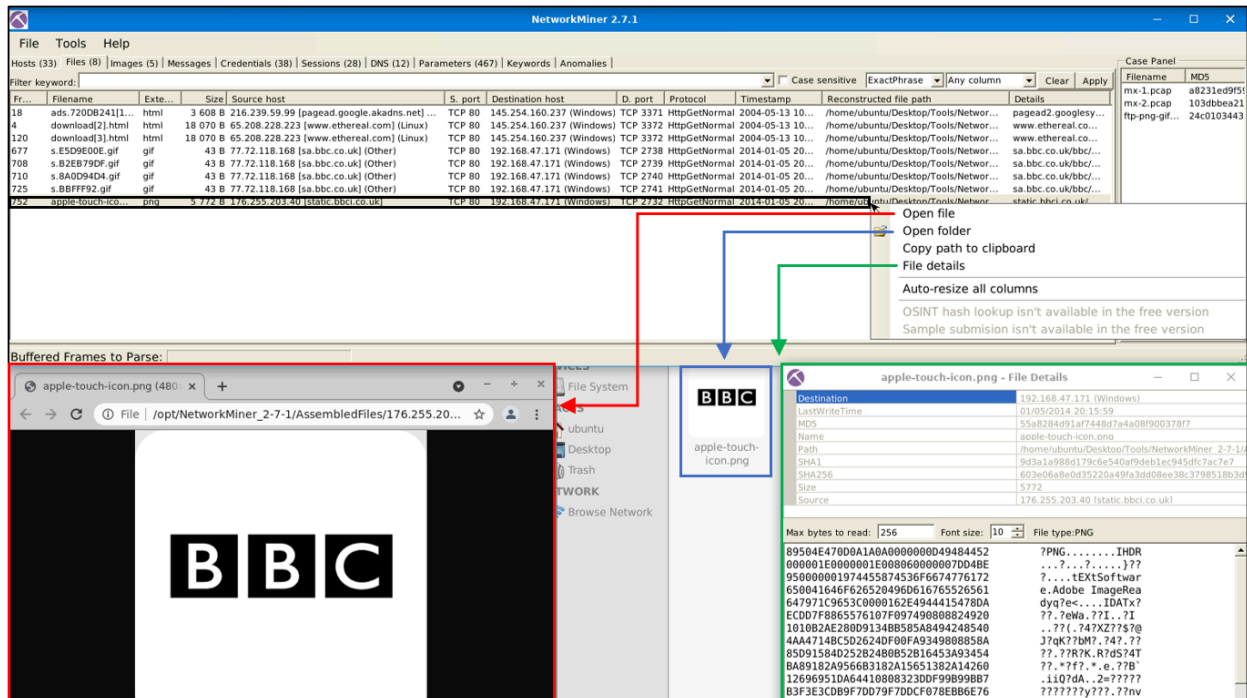
NETNTLMv2

#B\$136B077D942D9A63\$FBFF3C253926907AAAAD670A9037F2A5\$0101000000000000094D71AE38CD60170A8D57112

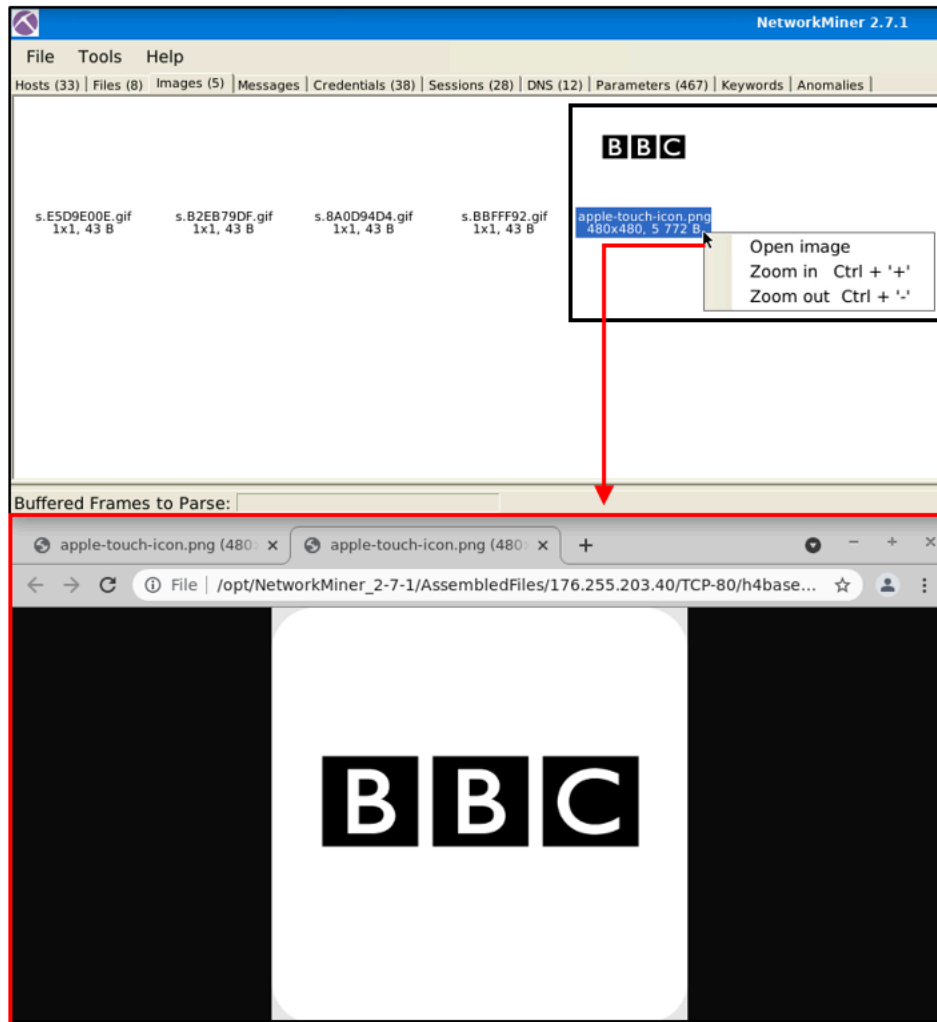
Task 5 - Tool Overview 2

Files: The file menu shows extracted files from investigated pcaps. This section provides information on:

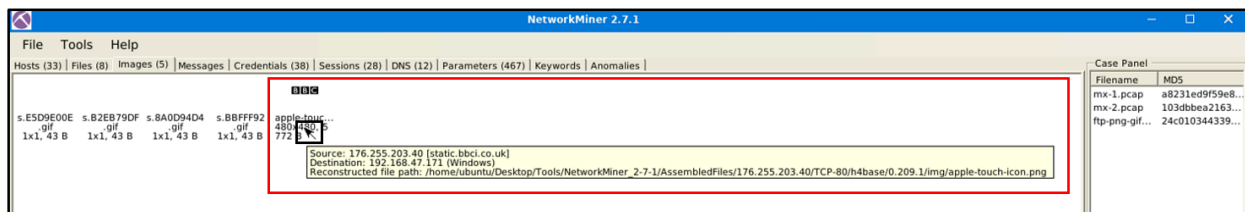
- Frame number
- Filename
- Extension
- Size
- Source and destination address
- Source and destination port
- Protocol
- Timestamp
- Reconstructed path
- Details



Images: The file menu shows extracted images from investigated pcaps. The right-click menu is helpful in this part as well.



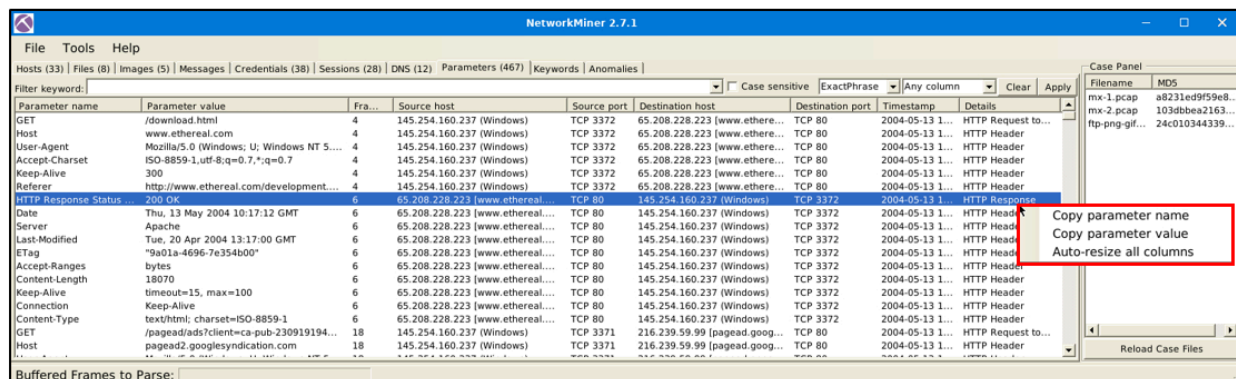
Once you hover over the image, it shows the file's detailed information (source & destination address and file path).



Parameters: The file menu shows extracted parameters from investigated pcaps. This section provides information on:

- Parameter name
- Parameter value
- Frame number
- Source and destination host
- Source and destination port
- Timestamp
- Details

The right-click menu is helpful in this part as well. You can copy the parameters and values easily.



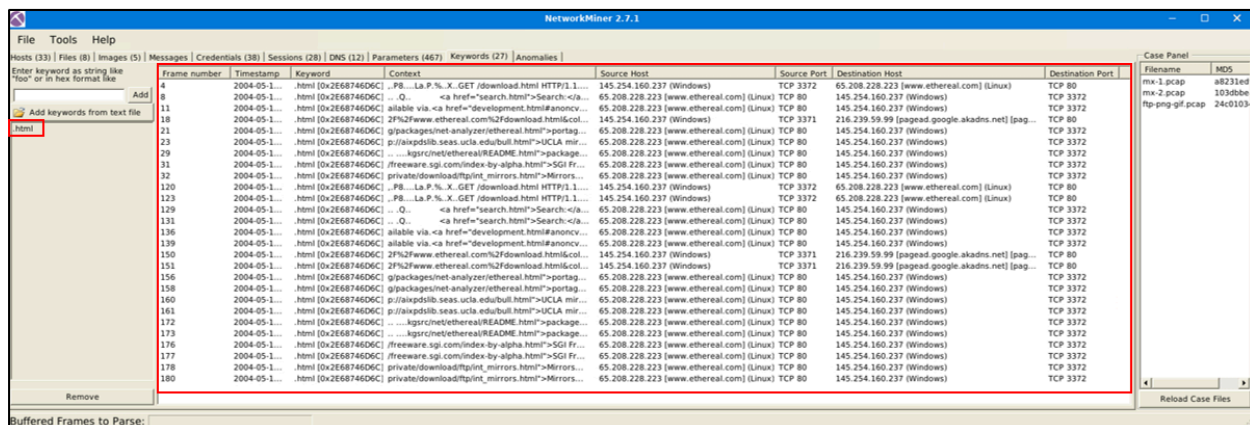
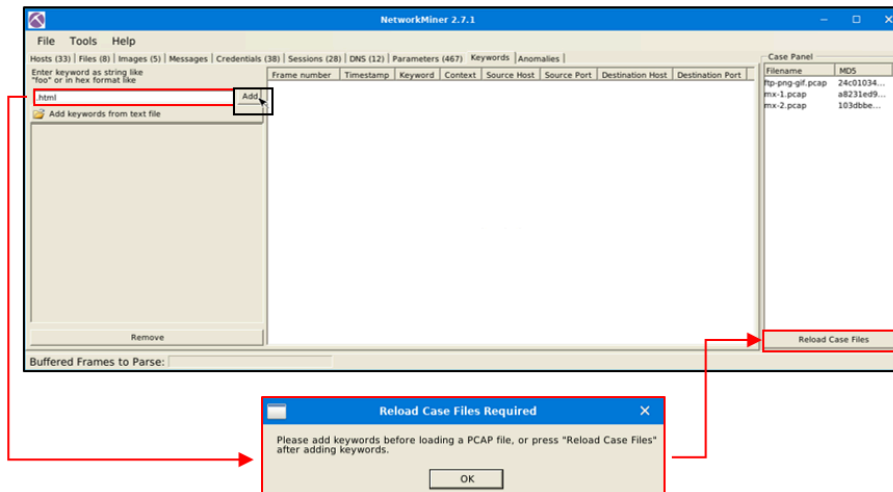
Keywords: The file menu shows extracted keywords from investigated pcaps. This section provides information on:

- Frame number
- Timestamp
- Keyword
- Context
- Source and destination host
- source and destination port

How to filter keywords:

- Add keywords
- Reload case files!

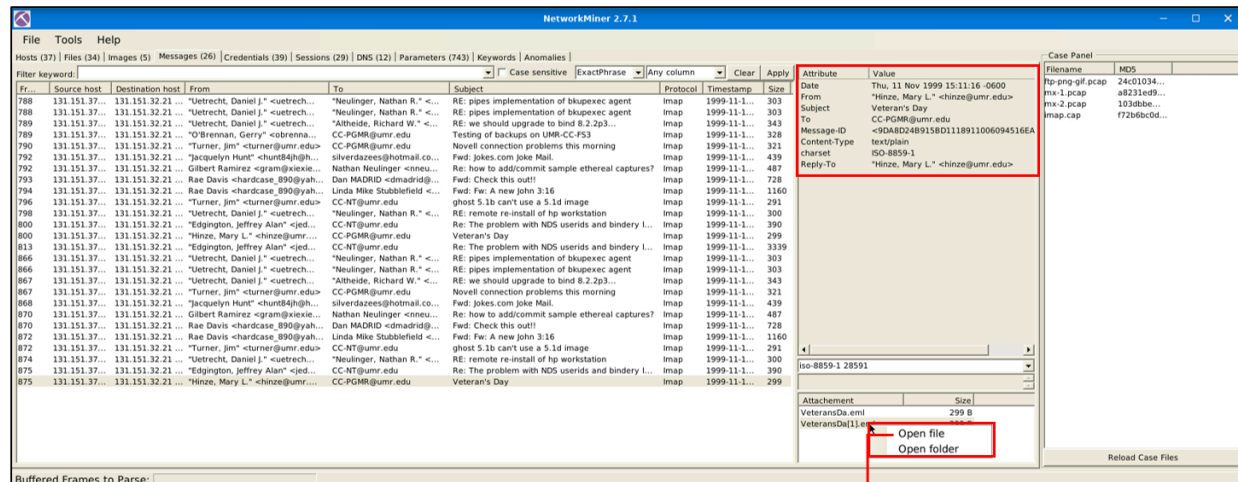
Note: You can filter multiple keywords in this section; however, you must reload the case files after updating the search keywords. Keyword search investigates all possible data in the processed pcaps.



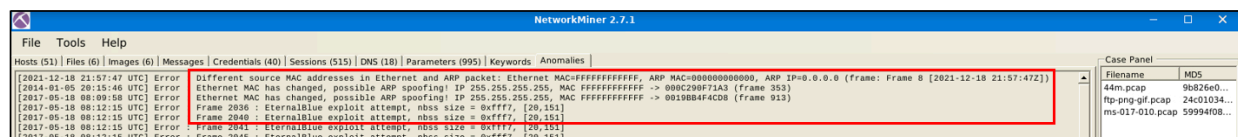
Messages: The messages menu shows extracted emails, chats and messages from investigated pcaps. This section provides information on:

- Frame number
- Source and destination host
- Protocol
- Sender (From)
- Receiver (To)
- Timestamp
- Size

You can use the built-in viewer to investigate overall information and the "open file" option to explore attachments.



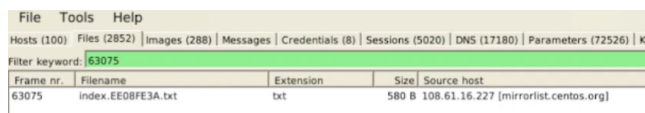
Anomalies: The anomalies menu shows detected anomalies in the processed pcap. Note that NetworkMiner isn't designated as an IDS. However, developers added some detections for EternalBlue exploit and spoofing attempts.



Answer the questions below

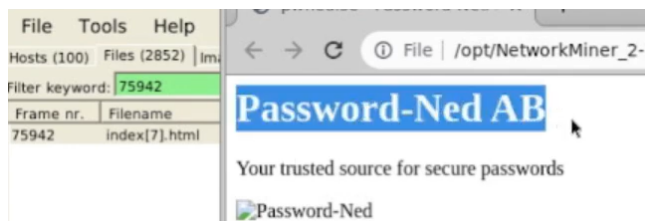
Use "~/Desktop/Exercise files/mx-7 pcap" file to answer the questions below. What is the name of the Linux distro mentioned in the file associated with frame 63602?

(Note: If there are no results associated with this frame, check out frame 63075) CentOS



What name and surname are mentioned in the file associated with frame 76469?

(Note: If there are no results associated with this frame, check out frame 75942) Ned Flanders (right click open file)



What is the source address of the image ads.bmp.2E5F0FD9[1].bmp"? 80.239.178.187

Hosts (100)	Files (2852)	Images (288)	Messages	Credentials (8)	Sessions (5020)	DNS (17180)	Parameters (72526)	Keywords	Anomalies
Filter keyword:	ads.bmp.2E5F0F09.bmp								
Frame nr.	Filename	Extension	Size	Source host	S. port	Destination host	D. port	P.	
82674	ads.bmp.2E5F0F09.bmp	bmp	126 B	80.239.18.187 [a51.dscg10.akamai.net] [a49.r...	TCP 80	192.168.0.51 [homer-xubuntu] (Linux)	TCP 43717	H	

What is the frame number of the possible TLS anomaly? 36255

Messages	Credentials (8)	Sessions (5020)	DNS (17180)	Parameters (72526)	Keywords	Anomalies
TLS data boundary is not on a TLS record boundary in frame 36255 TLS data boundary is not on a TLS record boundary in frame 73673						

Use "~/Desktop/Exercise files/mx-9" file to answer the questions below. Which platform sent an email with the subject starting with "You have more...?"

Facebook (Click on the **Messages** section.)

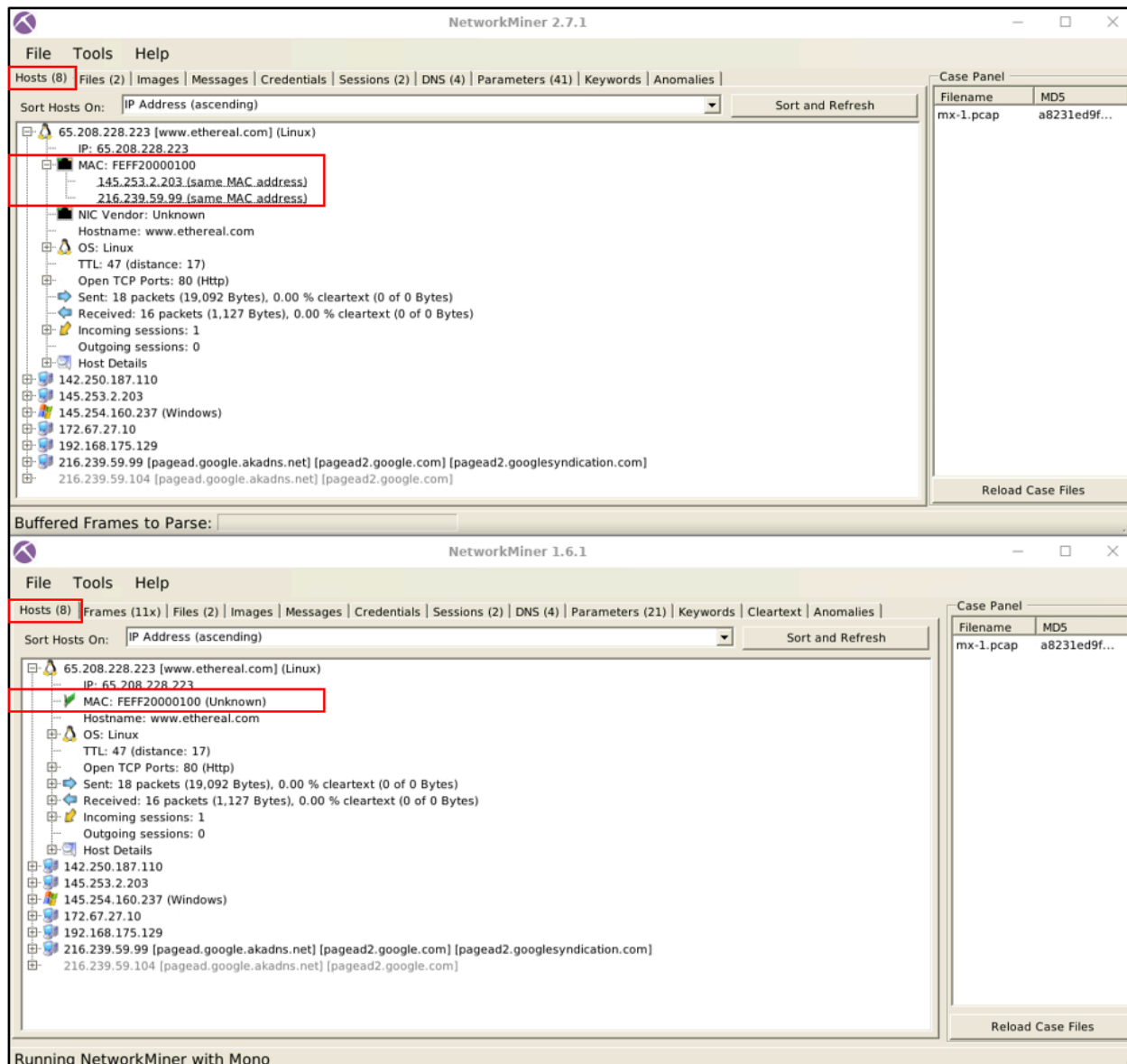
File	Tools	Help
Hosts (830)	Files (854)	Images (50)
Messages (12)	Credentials (74)	Sessions (2416)
DNS (9583)	Parameters (34842)	Keywords
Anomalies		
Filter keyword:		
	Destination host	From
		To
171 [pop.gmx.com]	192.168.0.51 [HOMER-XUBUNTU<00>] [HOMER...	Beyond Security Support <support@beyondsecu...> homer.pwned.se@gmx.com
187 [pop.gmx.com]	192.168.0.51 [PASSWORD-NEO-XP<20>] [PAS...	"Facebook" <update+zrdzhehcvf1@facebookma...> Password-fled Flanders <ned.pwne
187 [pop.gmx.com]	192.168.0.51 [HOMER-XUBUNTU<00>] [HOMER...	Branson Matheson <branson@sandsite.org> LOPSA Discuss List <discuss@lopsa
187 [pop.gmx.com]	192.168.0.51 [HOMER-XUBUNTU<00>] [HOMER...	Branson Matheson <branson@sandsite.org> LOPSA Discuss List <discuss@lopsa
187 [pop.gmx.com]	192.168.0.51 [HOMER-XUBUNTU<00>] [HOMER...	Matt Finnigan <mfinnigan@gmail.com> Branson Matheson <branson@sanc

What is the email address of Branson Matheson? branson@sandsite.org

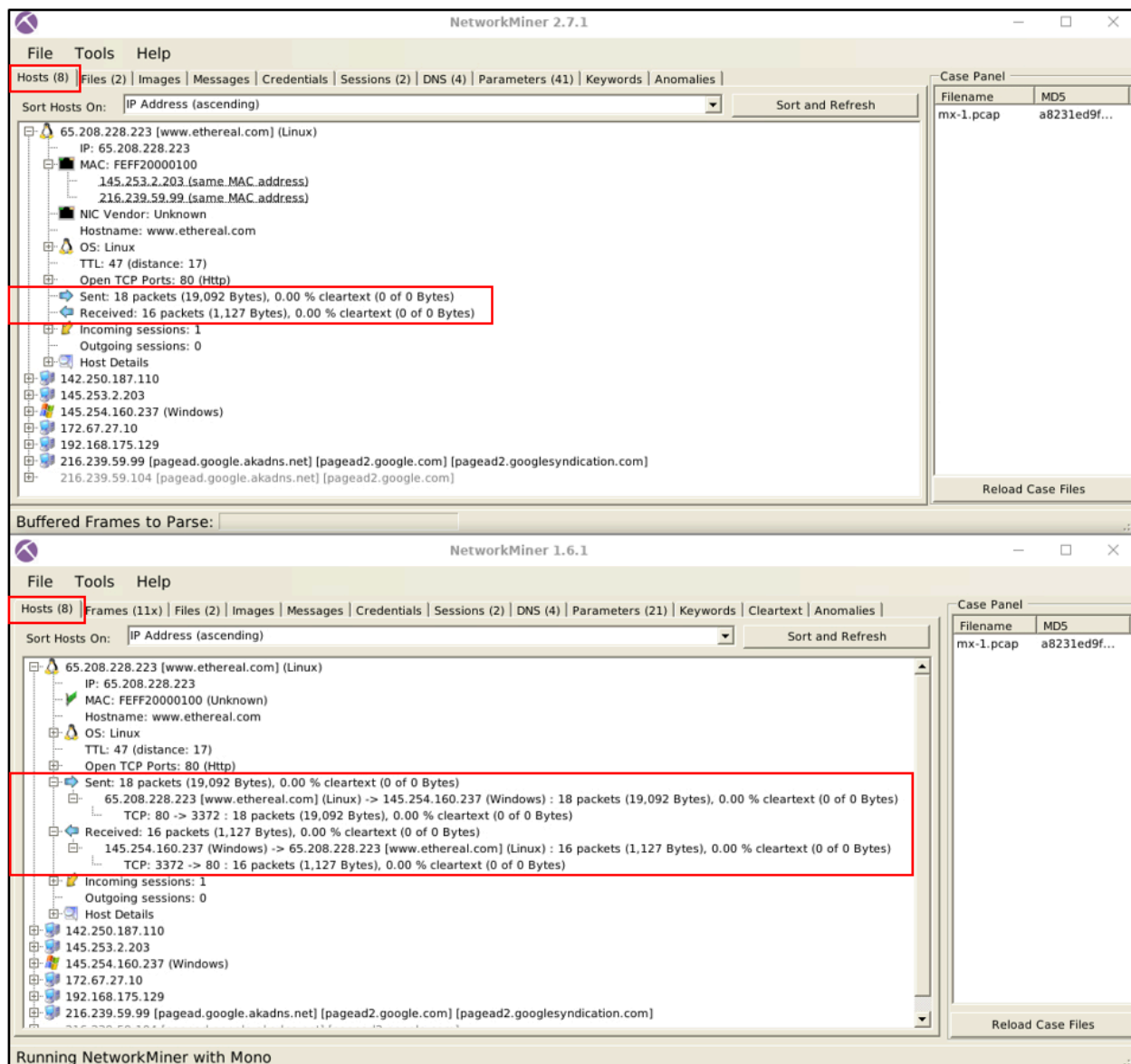
Task 6 - Version Differences

Version Differences: Newer NetworkMiner versions improve stability, security, and features, with v2.7 offering significant enhancements over v1.6.

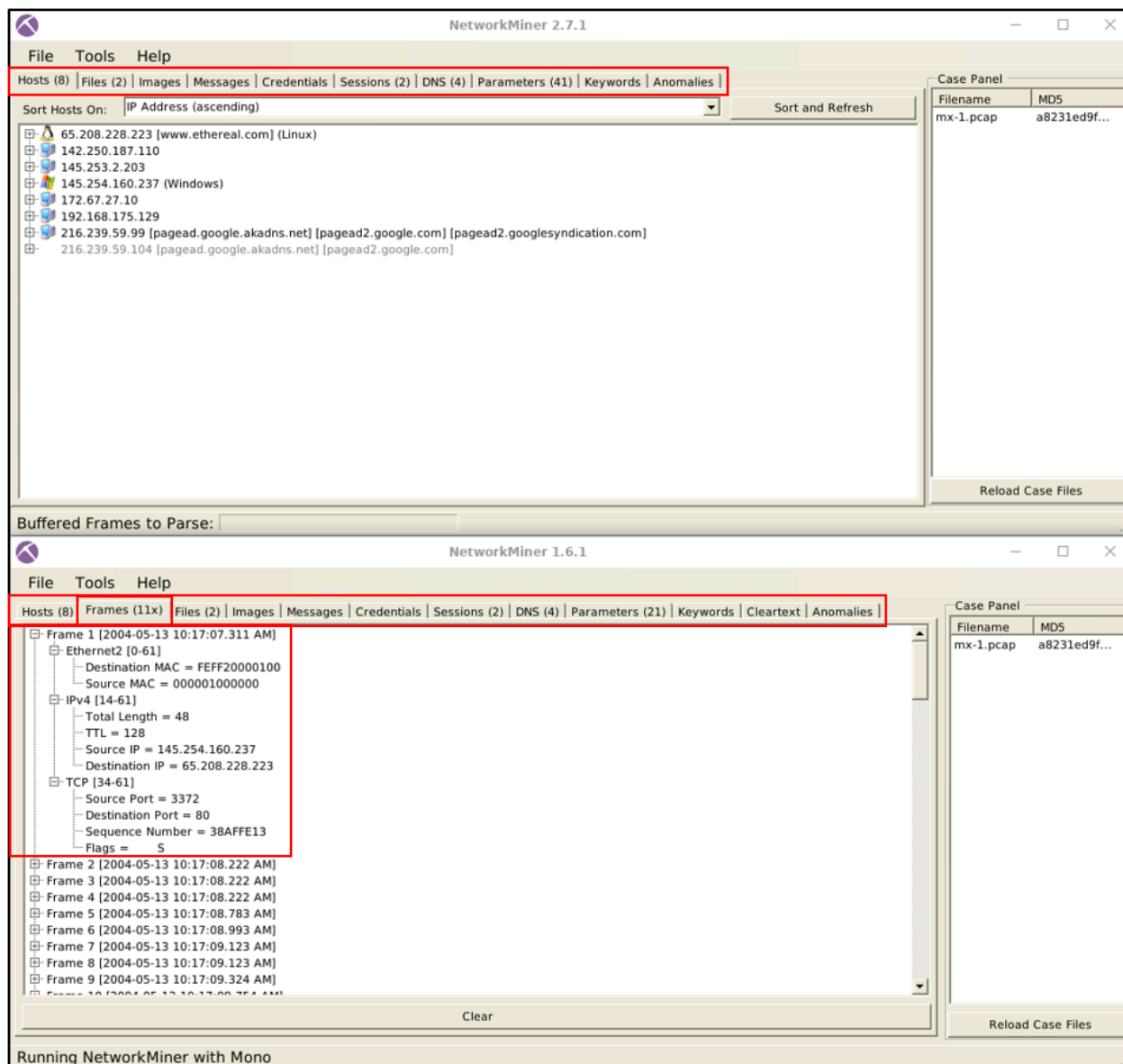
Mac Address Processing: This option will help you identify if there is a MAC Address conflict. This feature is not available before version 2.



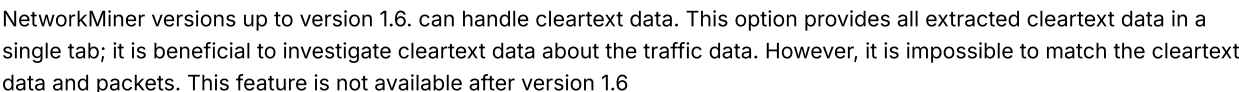
NetworkMiner versions up to version 1.6. can handle packets in much detail.

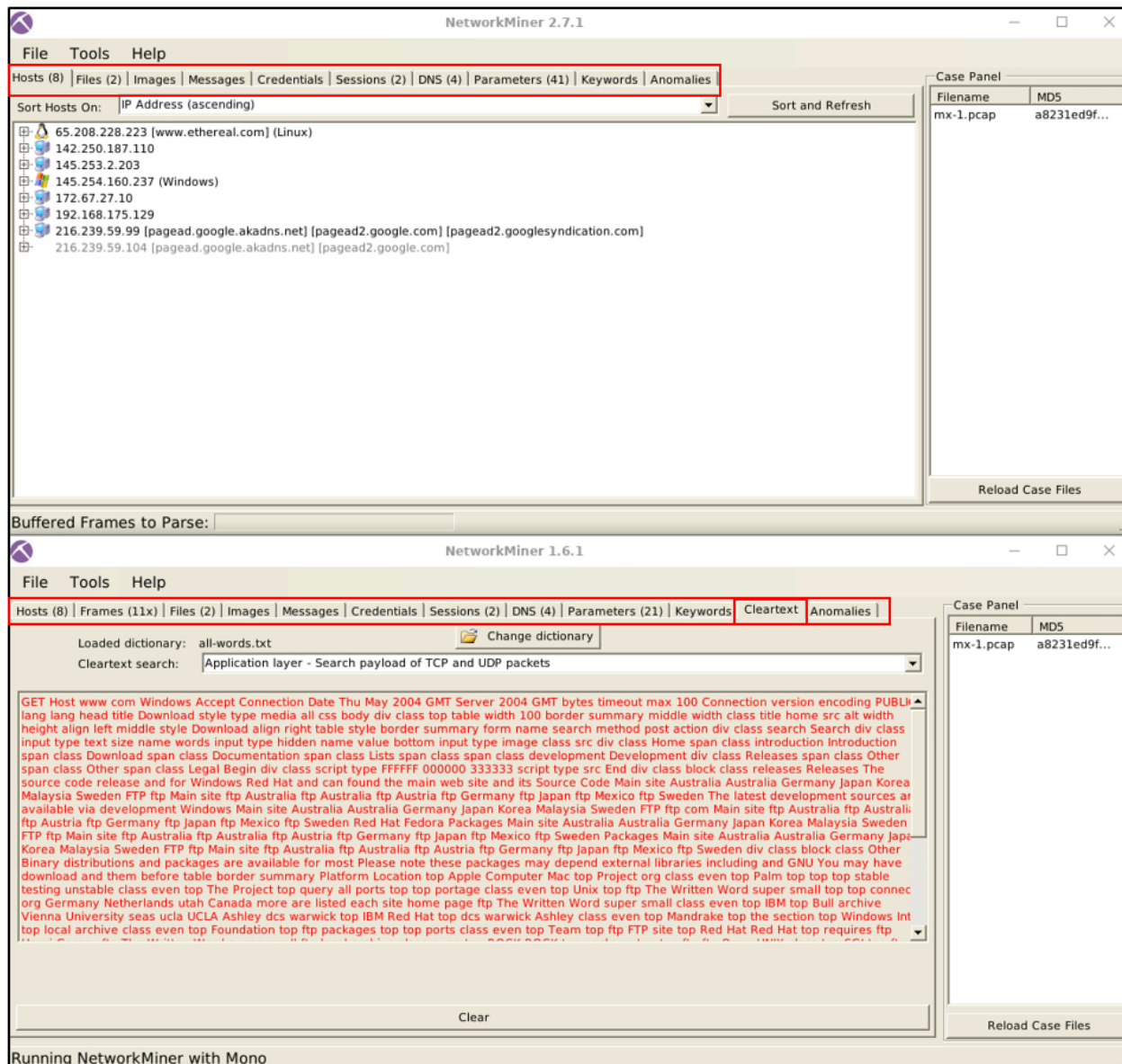


Frame Processing: NetworkMiner versions up to version 1.6. can handle frames. This option provides the number of frames and essential details about the frames. This feature is not available after version 1.6.



Parameter Processing: NetworkMiner versions after version 2 can handle parameters in a much more extensive form. Therefore version 1.6.xx catches fewer parameters than version 2.





Answer the questions below

- | Which version can detect duplicate MAC addresses? 2.7
- | Which version can handle frames? 1.6
- | Which version can provide more details on packet details? 1.6

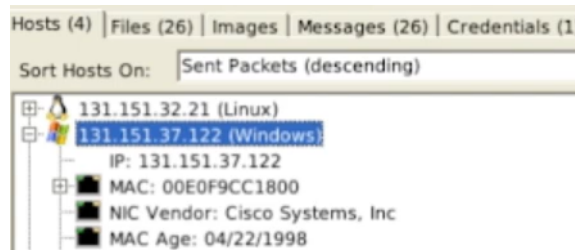
Task 7 - Exercises

The upload may take up to 5 minutes, and NetworkMiner might look unresponsive for some time. For better performance, close NetworkMiner, then open it again before uploading the case1 PCAP file, and then again for the case2 PCAP file.

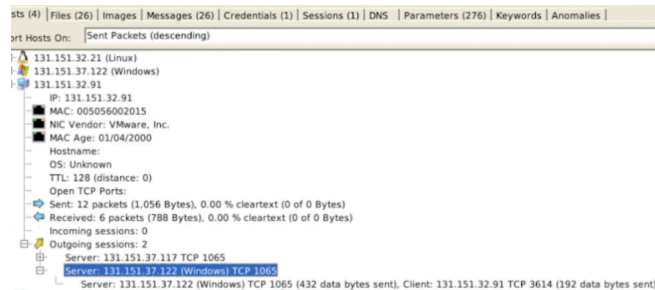
Answer the questions below

- | Use the ~/Desktop/Exercise Files/case1.pcap file to answer the questions below. What is the full OS name of the host 131.151.37.122?

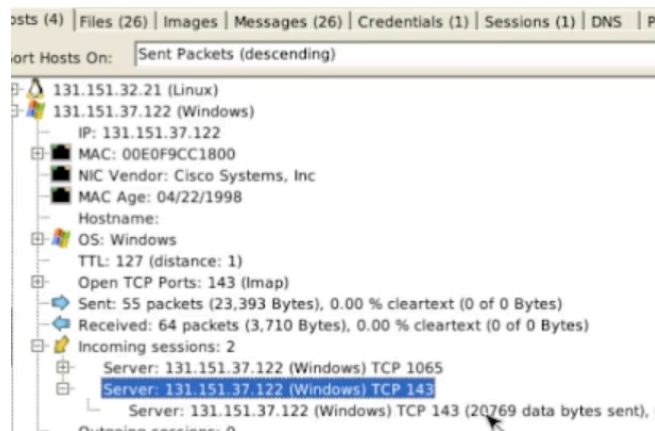
Windows - Windows NT 4 (click on host section, select IP)



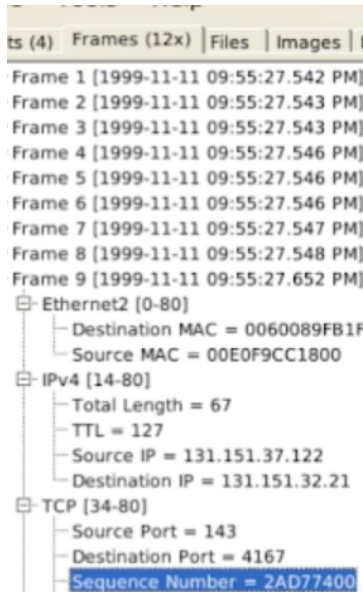
Investigate the hobetween 131.151.37.122 and 131.151.32.91. How many bytes were sent by the client (*.32.91) through port 1065? 192



Investigate the communication between 131.151.37.122 and 131.151.32.21. How many bytes were sent back by the server (*.37.122) through port 143? 20769



What is the sequence number of frame number 9? 2AD77400



What is the number of the detected "content types"?

2 (Click **Parameters**. There are two different content types: **multipart/mixed** and **text/plain**.)

Use the ~/Desktop/Exercise Files/case2.pcap file to answer the questions below. What is the USB product's brand name?
ASIX

(472)	Files (2190)	Images (612)	Messages	Credentials (312)	Sessions (2828)	DNS (8940)	Parameters (55)
keyword:	usb						
se nr.	Filename	Extension	Size	Source host			
1	hi-Speed-usb2.0_ax88772.htm	html	15 827 B	210.243.224.51 [asix.com.tw]			

What is the name of the phone model? Lumia 535 (click on file, filter phone)

What is the source IP of the fish image? 50.22.95.9 (click on file, filter fish)

What is the password of the homer.pwned.se@gmx.com? spring2015 (click on credentials, filter email)

What is the DNS query of frame 62001? pop.gmx.com (click on dns, filter frame)

Task 8 - Conclusion

In this room, we covered NetworkMiner, what it is, how it operates, and how to investigate pcap files. As I mentioned in the tasks before, there are a few things to remember about the NetworkMiner:

- Don't use this tool as a primary sniffer.
- Use this tool to overview the traffic, then move forward with Wireshark and tcpdump for a more in-depth investigation.